UGS v2 — Universal Generation Specification

Status: v2.0 (draft specification)
Origin: Derived from the "UGS v2 — Design Goals" design note by Konstantin Parashchuk.
Purpose: A model-agnostic specification for generating images and video from *world simulation* rather than *picture description*.

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0. What UGS Is — and Is Not

UGS is **not a prompt**.

UGS is a **specification of a virtual reality**.

Any image or video is a *consequence* of a simulation, never its goal. The generator must never ask itself:

> "How do I make a beautiful shot?"

It must ask:

> "What frame would naturally arise in this world?"

This single inversion — goal becomes byproduct — is the defining property of UGS and the reason it differs from "cinematic prompt" collections that merely accumulate visual tricks. UGS defines a **decision hierarchy**: first the construction of the world and its causality, then the modeling of its state, and only at the very end the selection of the observed image or video.

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1. Priority Order (Non-Negotiable)

Always respect the following order. It is the spine of the entire system.

```
Reality
↓
Simulation
↓
Observation
↓
Rendering
```

Never the inverted, decorative pipeline:

```
Prompt
↓
Composition
↓
Beautiful Image
```

The first chain is causal and bottom-up. The second is decorative and top-down. UGS forbids the second.

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2. The Master Principle

> **Do not describe the picture. Describe the world. The picture emerges on its own.**

Everything in UGS is a consequence. A frame is not composed; it is *witnessed*.

Do not write	Write instead
A man stands.	A man has been walking through the forest for several hours. The camera happened to see him just now.
A destroyed house.	A house destroyed by artillery three days ago.
Rain.	An atmospheric front passed during the night.

The verb tense matters. States are described as *outcomes of prior events*, not as static labels.

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3. Everything Has a Past

Every object carries:

- **origin** — where it came from
- **history** — what happened to it
- **age** — how long it has existed in this state
- **owner** — who is responsible for it
- **state** — its current condition, including wear
- **future** — what will become of it

Nothing appears *ex nihilo*. A detail without a backstory is a detail that will read as artificial.

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4. The World Exists Outside the Frame

This is the most important rule.

> A frame shows less than 1% of the world. Everything else keeps living.

Examples of the implied exterior:

- If there is a train — it is going **somewhere**.
- If there is a soldier — he has **received an order**.
- If a hatch is open — **someone** opened it.
- If a lamp is lit — the **power plant is running**.

The off-screen world is not decoration. It is the cause of what is on screen.

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5. Causality

Nothing exists without a reason. For every object the agent must mentally answer:

- Why did it appear?
- Who made it?
- When?
- Why **here** specifically?
- What will happen in one minute?

If any of these cannot be answered, the object is likely a decorative artifact and should be removed or given a cause.

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6. Materials

Every material has:

- **behavior** under stress and time
- **wear** pattern
- **history** of use
- **physics** of its response

Examples:

- Wood ages differently from concrete.
- Concrete ages differently from plastic.
- Steel ages differently from copper.

Material mismatch (e.g., a "new" object in an ancient ruin) is a causality violation.

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7. Light

Light is not painted. It **propagates**.

Every source has:

- **power**
- **temperature** (color)
- **scattering**
- **reflections**
- **falloff / attenuation**
- **atmosphere** it passes through

Describe illumination as a physical emission-and-transport problem, not as a mood word.

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8. Camera

The camera is an **observer**, not a director. It emphasizes nothing. It happened to be here by chance.

8.1 Optics — Hyperfocal Focus (default)

By default:

> The whole world is equally worthy of attention.

There is no "main subject." There is a **main idea**.

Never apply, unless part of the story itself:

- bokeh
- portrait blur
- artificial depth of field

8.2 Composition

Composition arises by itself. It cannot be "drawn." Beauty is a *consequence of the world's structure*.

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9. Reference Synthesis

If references are given:

> Never copy frames. Extract only the *language*.

Extract, then build a new world:

- language of form
- language of material
- language of light
- language of camera
- language of architecture
- language of color
- language of editing
- language of animation

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10. Identity Preservation

If a photograph of a person is given:

- Preserve the person first.
- Then change the style.
- Never the reverse.

Only the **visual language** changes. Never the **personality / identity**.

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11. Environmental Storytelling

Every corner of the frame tells a story. Even trash. Even a scratch. Even an open door.

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12. Temporal Continuity

Video is **not a sequence of frames**. It is a continuous physical process. Every motion has:

- acceleration
- inertia
- mass
- a beginning moment
- an ending moment

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13. Animation

Characters never *act*. They **live**. All movement must have a purpose. No "beautiful gestures" that are caused by nothing.

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14. Audio (for video)

Sound is part of physics, not an ornament.

- If an engine exists — it makes noise.
- If wind is strong — it affects the sound.
- If a room is large — there is reverberation.

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15. Negative Constraints

Never produce:

- random surrealism
- decorative details
- beautiful poses
- random effects
- meaningless chaos
- visual noise

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16. Memes

Absurdity is born not from randomness, but from **one single altered axiom of the world**. After that, the entire world develops logically around it.

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17. Horror

Monsters may not exist at all. Fear must arise from **understanding the world**.

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18. Games

Do not draw a screenshot. **Simulate an existing game engine.** Define:

- renderer
- GPU limits
- materials
- LOD
- shadows
- lighting
- textures
- hardware era

Then obtain a natural frame.

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19. Video (general)

Video must feel like a **continuous simulation**, not a sequence of beautiful frames.

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20. The Master Formula of UGS

The most important rule reduces to one chain:

```
Ontology
↓
World Rules
↓
History
↓
Simulation
↓
Physics
↓
Environment
↓
Characters
↓
Camera
↓
Observation
↓
Image / Video
```

This is the decision hierarchy that distinguishes UGS from picture-centric prompting. Each level is a prerequisite for the next; skipping a level injects artificiality.

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21. Authoring Checklist (Operational)

Before emitting a generation request in UGS style, confirm:

- **Ontology** — what kinds of things exist in this world?
- **World Rules** — what are the invariant laws (physics, society, tech)?
- **History** — what happened before the frame? (origin, age, owner)
- **Simulation** — what state is the world in *right now*, off-screen included?
- **Physics** — light, material, mass, inertia modeled as real?
- **Environment** — does the off-frame world have a cause for the on-frame content?
- **Characters** — do they have purposes, not poses?
- **Camera** — observer by chance, hyperfocal, no artificial DOF?
- **Observation** — is the frame a *witnessed* slice, not a composed one?
- **Image/Video** — does the output *emerge* rather than *decorate*?

If any level is missing or contradicts an earlier one, return to that level. Do not patch at the rendering stage.

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22. Worked Example

Bad (picture-centric): "Cinematic shot of a tired soldier standing in a ruined city, dramatic lighting, shallow depth of field."

UGS (world-centric):

> *Ontology:* A mid-20th-century continental war. Humans, machinery, municipalities.
> *World Rules:* Artillery outranges infantry movement; supply lines determine where units halt.
> *History:* The district was shelled on the third of the month; the building behind the figure lost its eastern wall then.
> *Simulation:* A conscript's platoon received a hold order at dawn. He has been stationary near the intersection for four hours awaiting relief.
> *Physics:* Overcast diffuse light, no sun; concrete dust settled on wool; brass casing oxidized where it landed.
> *Environment:* A supply truck is overdue from the west — its absence is why he is still here.
> *Characters:* He is not posing. He is waiting, weight on one leg, ration cold.
> *Camera:* A fixed surveillance mount on the intact water tower, recording regardless of interest.
> *Observation:* The frame is one of 14,400 recorded that day. Nothing was staged.

The image that emerges from this world is automatically coherent. The "beauty" is a side effect of structure.

End of UGS v2 specification.